Inertial Particles and Turbulence

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ENSTA - Institut Polytechnique de Paris, Palaiseau, France

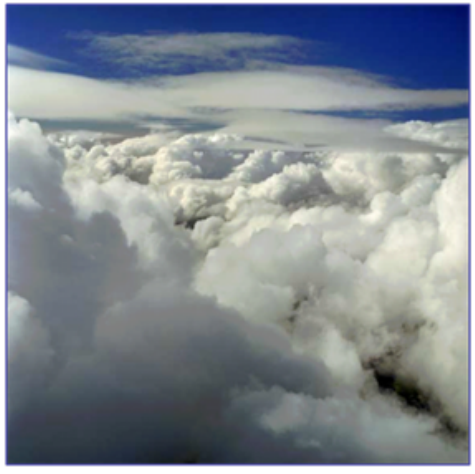


November 25th 2026 - Chatou

Inertial particles and turbulence

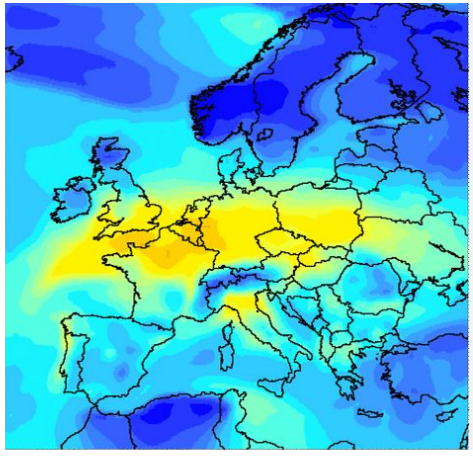










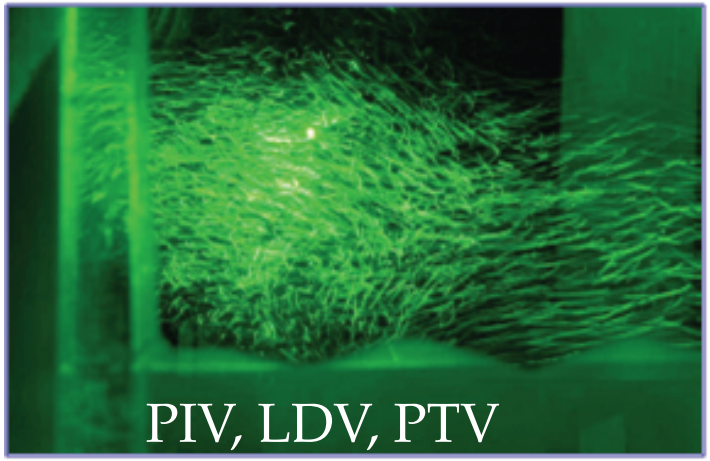




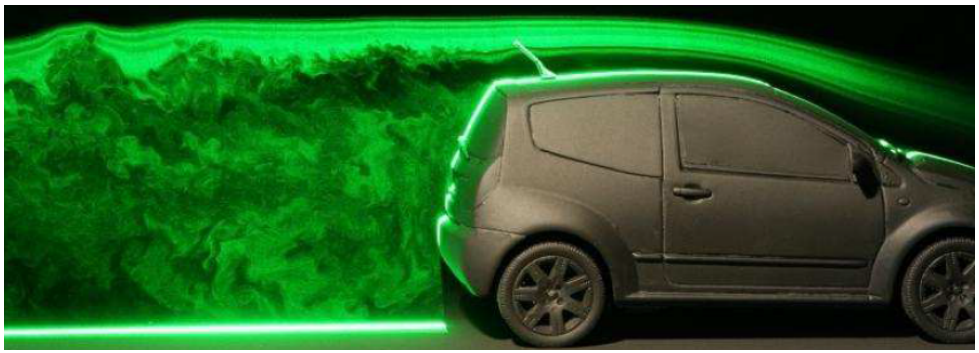


Tracers *vs* Inertial Particles

Tracers behave as fluid particles



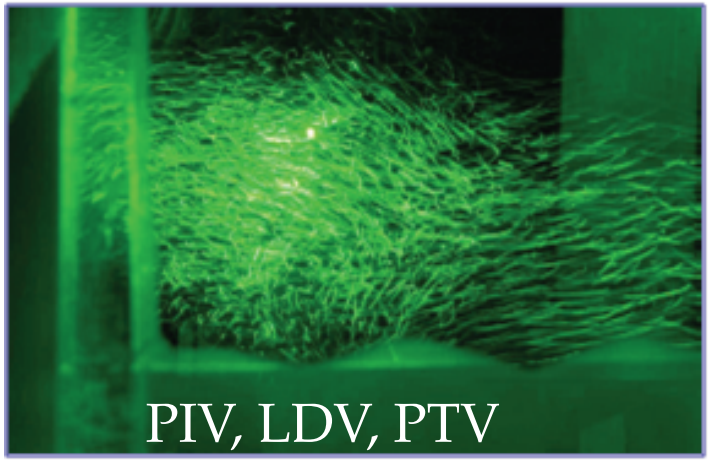
Inertial particles do not follow the flow



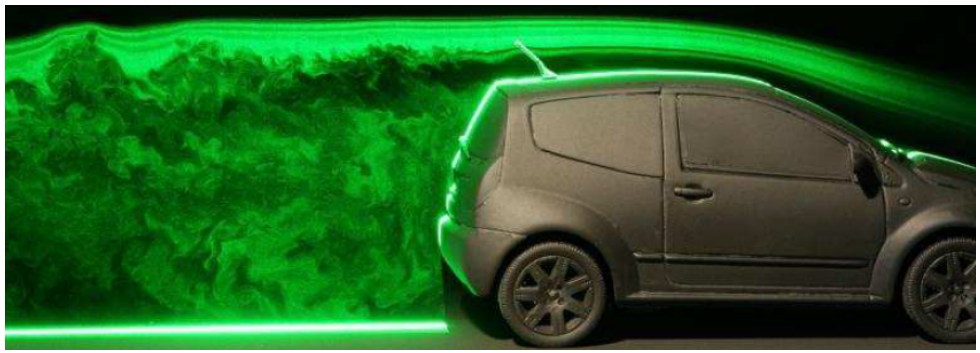
Grandemange et al, ENSTA 2016

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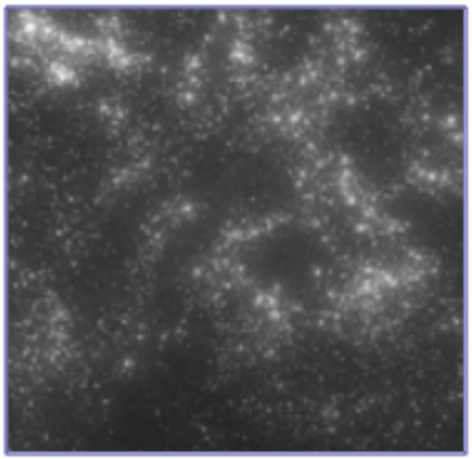


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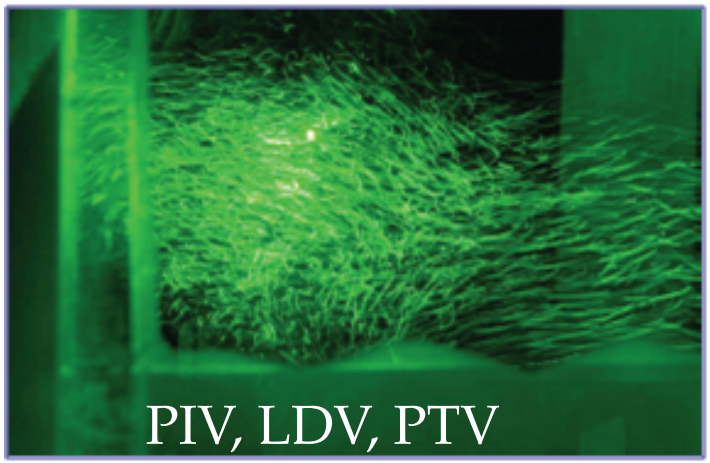
Cluster formation



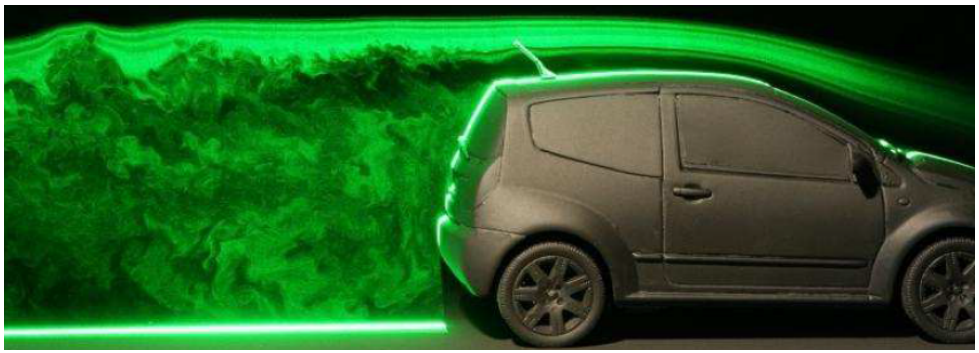
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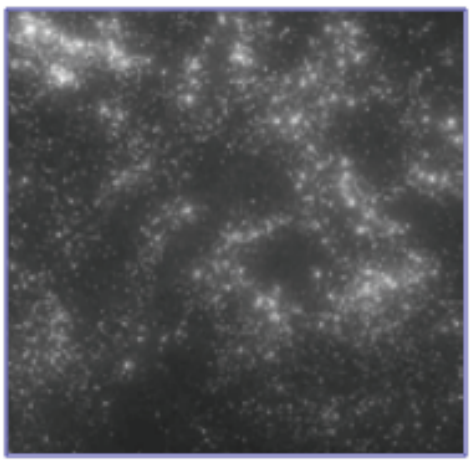


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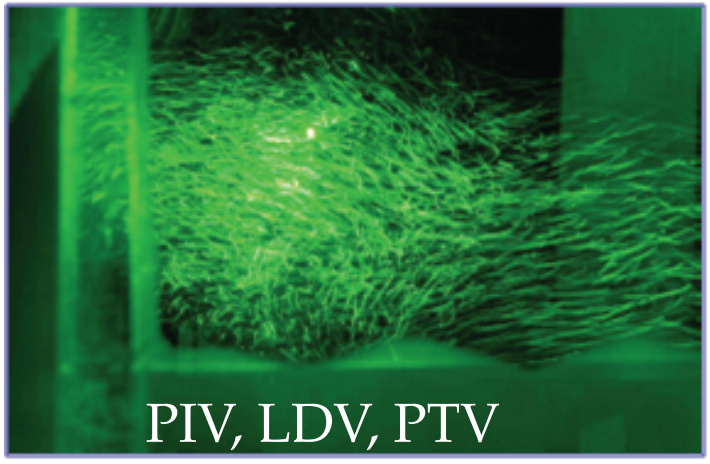
Settling alteration



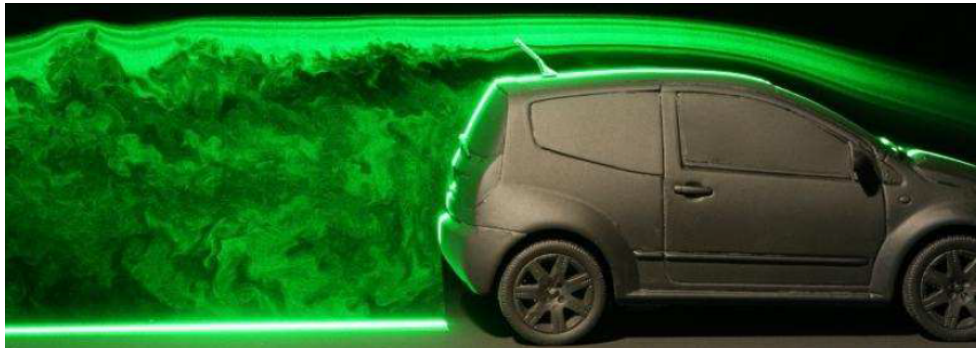
Artist view

Tracers *vs* Inertial Particles

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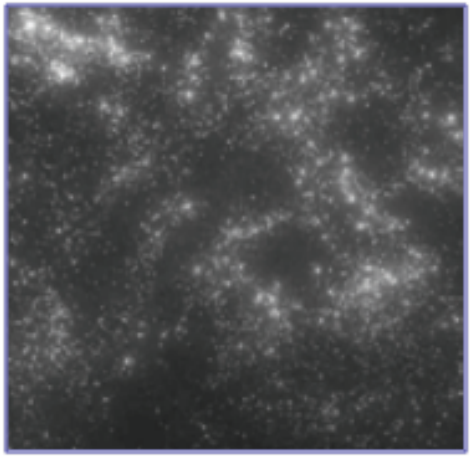


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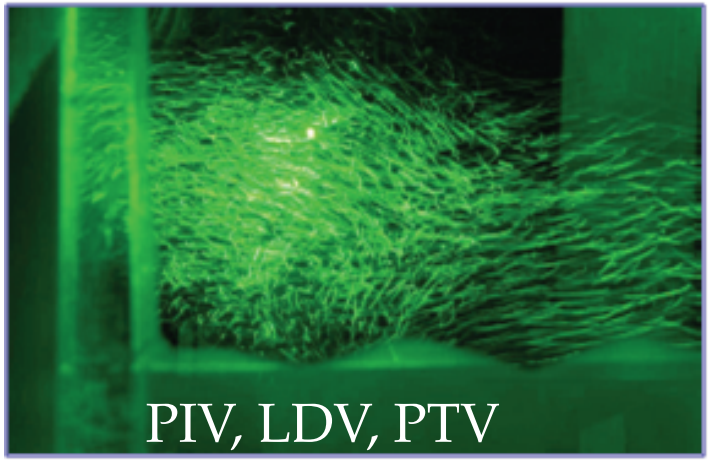
Collision alteration



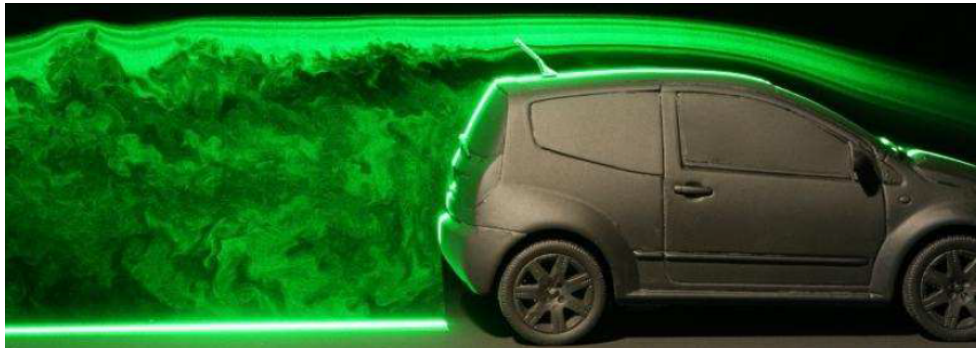
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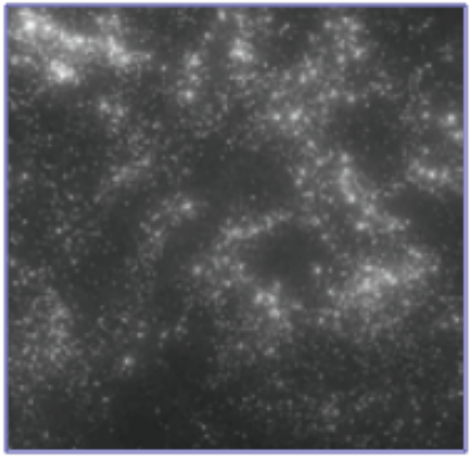


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Artist view

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Artist view

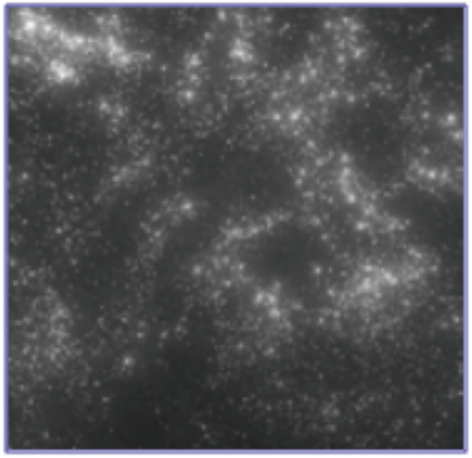
Flow modulation,
Dispersion...





Increase meeting probabilities

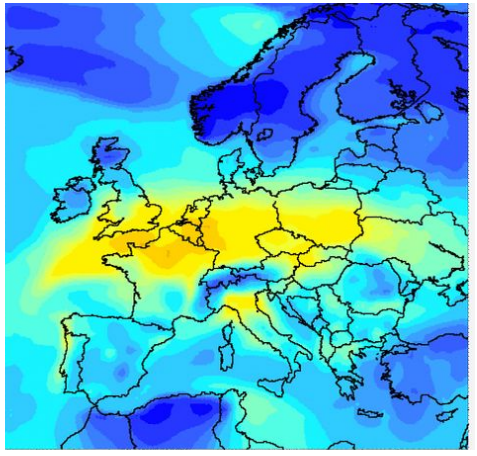
Cluster formation







Stoichiometric proportion



Sanitary thresholds







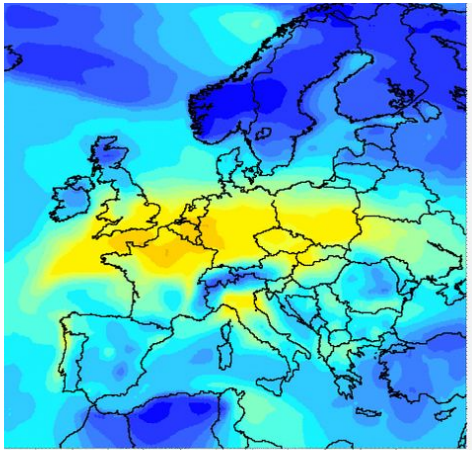
Settling alteration

-

Dispersion

Rain formation alteration

Aerosol and climate forcing
Aviation safety



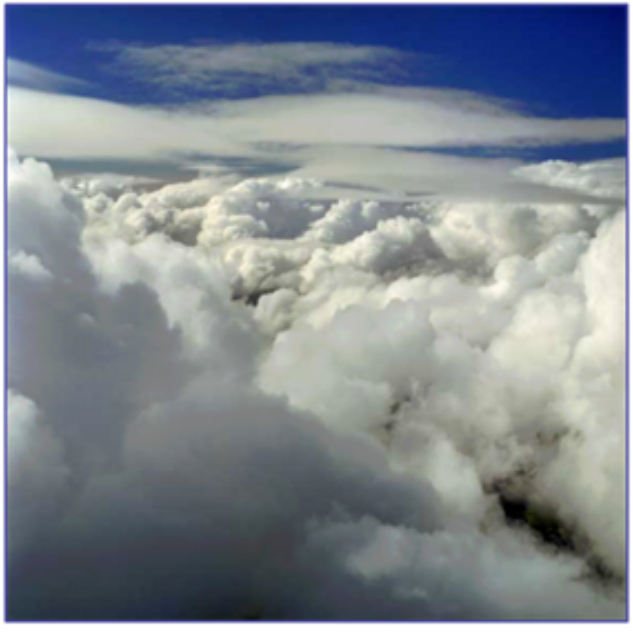




Marine snow : carbon pump

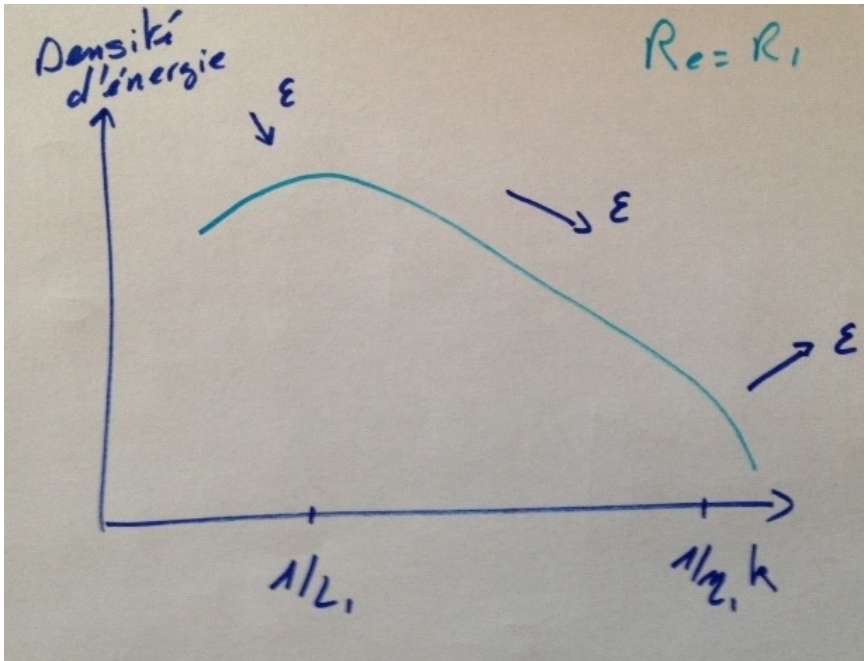
Estuaries dynamics

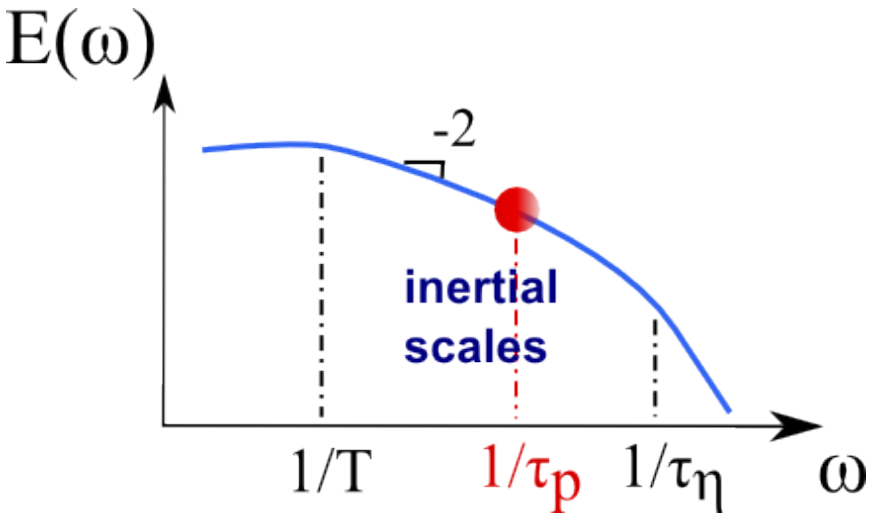


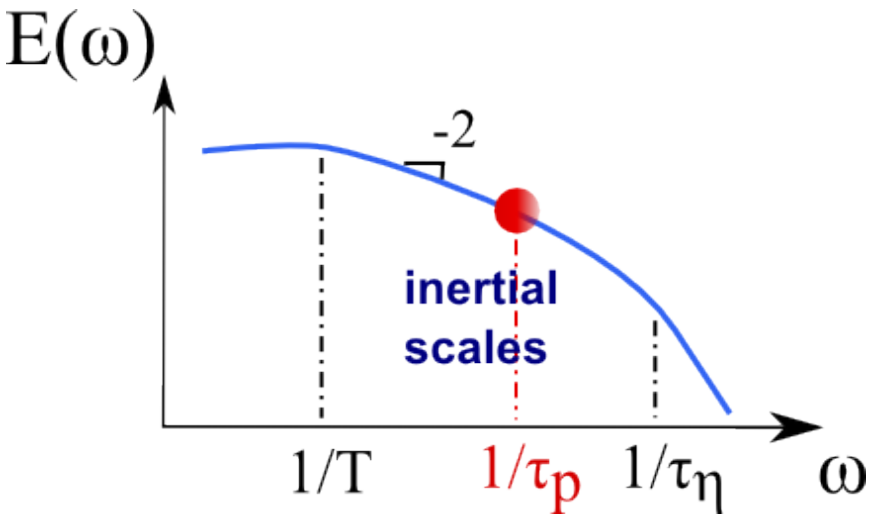







Densité d'énergie
ε
ε
ε
$Re = R_1$
$1/L_1$
$1/\eta_1 k$


E(ω)
-2
inertial
scales
1/T
1/τp
1/τη
ω

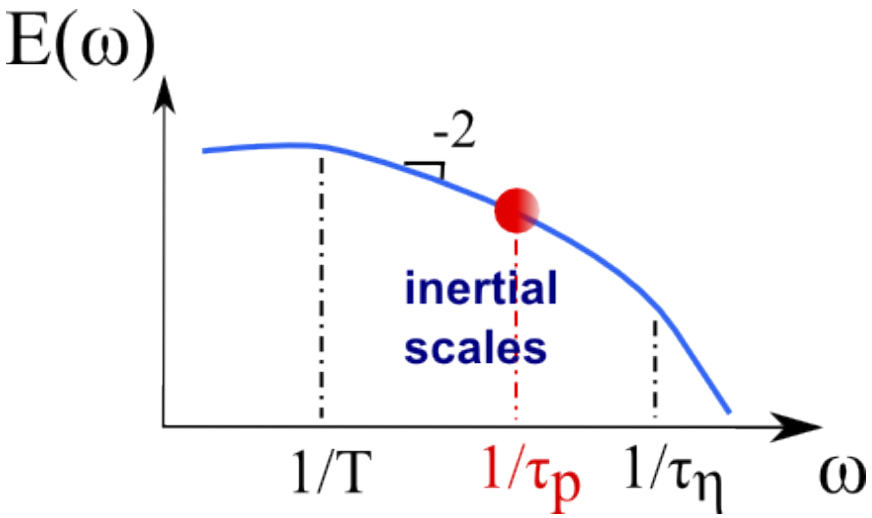


Model point particle equation :

$$\rho_p \frac{\partial v_i}{\partial t} = \overbrace{(\rho_p - \rho_f) g_i}^{\text{gravity}} + \overbrace{\frac{\partial \tau_{ij}}{\partial x_j} - \frac{\partial p}{\partial x_i}}^{\text{undisturbed flow}} + \overbrace{\frac{9}{2}\frac{\mu}{r_p^2}\left((u_i - v_i) + \frac{r_p^2}{6}\nabla^2 u_i\right)}^{\text{steady state drag}}$$

$$+ \overbrace{\frac{1}{2}\rho_f \frac{d}{dt}\left((u_i - v_i) + \frac{r_p^2}{10}\nabla^2 u_i\right)}^{\text{virtual or apparent mass term}} + \overbrace{\frac{24\mu}{r_p}\int_0^t \frac{d/dt\left(u_i - v_i + r_p^2/6\nabla^2 u_i\right)}{\pi\nu(t-\tau)^{1/2}} d\tau}^{\text{Basset history term}}$$

Maxey & Riley, *POF*, 26, **1983**
Gattignol, *JMTA*, 2, **1983**



Response time :

Stokes drag : $\tau_p = \rho_p d_p^2 / 18\mu$

Non linear drag : $\tau_p^{eff}(|\mathbf{w}|)$

Added mass : τ_p^{am}

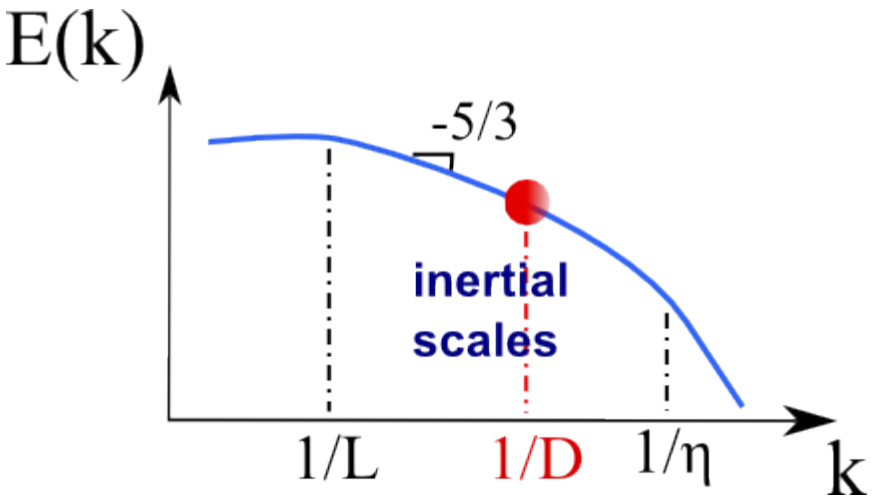
Basset history : ...

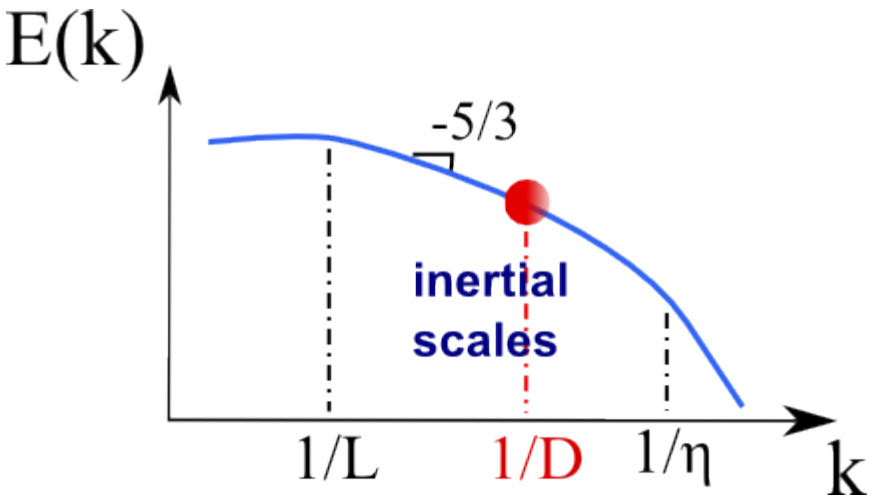
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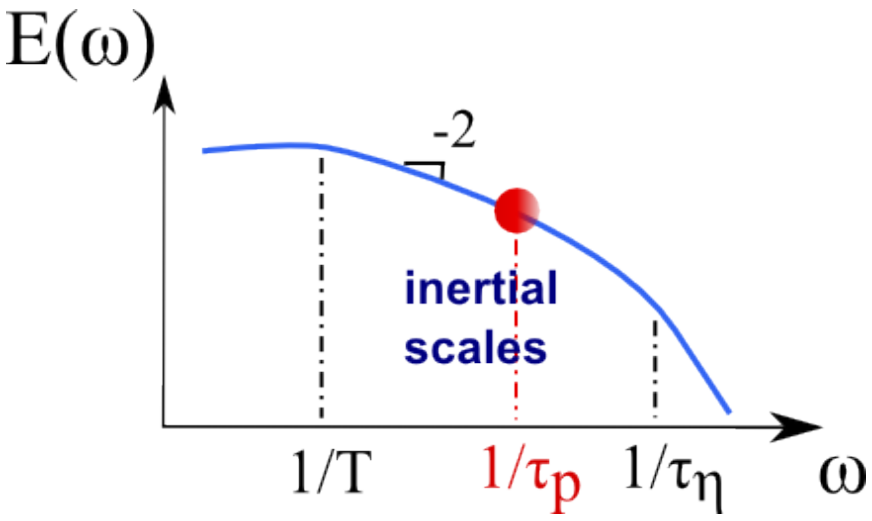
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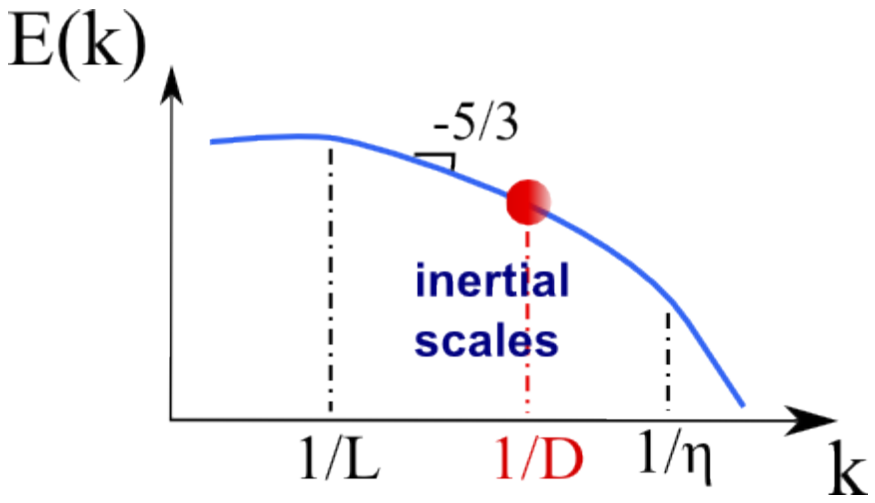

E(k)
-5/3
inertial
scales
1/L
1/D
1/η
k



Finite size effects :

- Faxén corrections ($d_p \leq \eta$)
- Filtering of small scales, integration over particle size
- Anisotropy
  - Rotation / tumbling inertia
  - Porosity and roughness
- …





Source	Timescale	Relevant when
Stokes drag	$\tau_p = \rho_p d_p^2/18\mu$	Always, baseline
Nonlinear drag	$\tau_p^{eff}(\lvert\mathbf{w}\rvert)$	$Re_p \gtrsim 1$
Added mass	τ_p^{am}	$\rho_p/\rho_f \sim \mathcal{O}(1)$
Basset history	complex kernel	Unsteady $\mathbf{w}$, ρ_p/ρ_f moderate
Orientational inertia	τ_{rot}	Non-spherical particles
Finite size filtering	d_p/η	$d_p \gtrsim \eta$
Porosity/permeability	Internal flow timescale	Aggregates, flocs

Control parameters

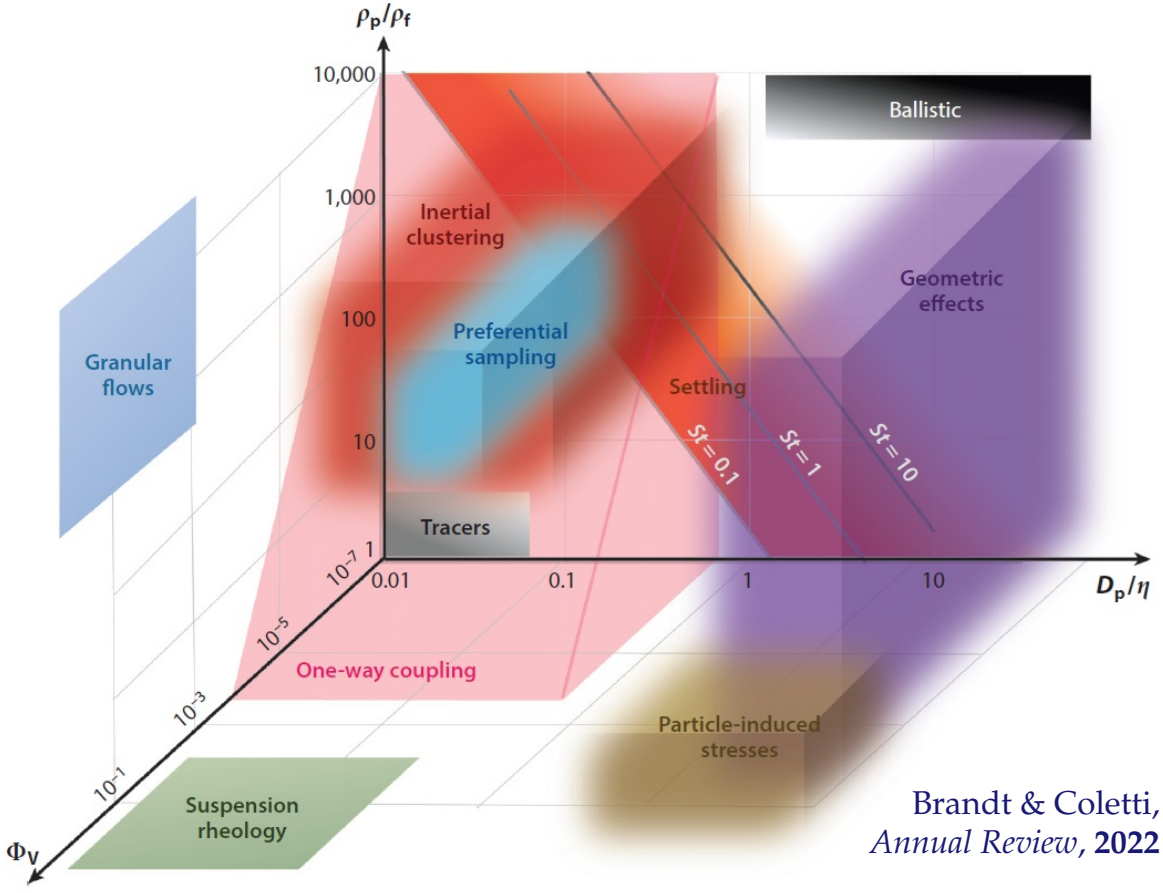
Reynolds number	Stokes number	Rouse number	Density ratio	Loading
$\mathbf{R}_\lambda = \frac{u'\lambda}{\nu}$;	$\mathbf{St} = \frac{2}{9}\left(\frac{r_p}{\eta}\right)^2\left(\frac{\rho_p}{\rho_f} - 1\right)$;	$\mathbf{R} = \frac{v_t}{u'}$;	$\Gamma = \frac{\rho_p}{\rho_f}$;	Φ_v.
Turbulence	**Inertia**	**Gravity**	**Density**	**Retroaction**
$10 - 10^4$	$10^{-5} - 10^2$	$0 - 10$	$10^{-4} - 10^3$	$10^{-6} - 10^{-1}$

- Particle Reynolds number
- Galileo number
- Aspect ratio
- Deformability
- ...

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Brandt & Coletti, *Annual Review*, **2022**

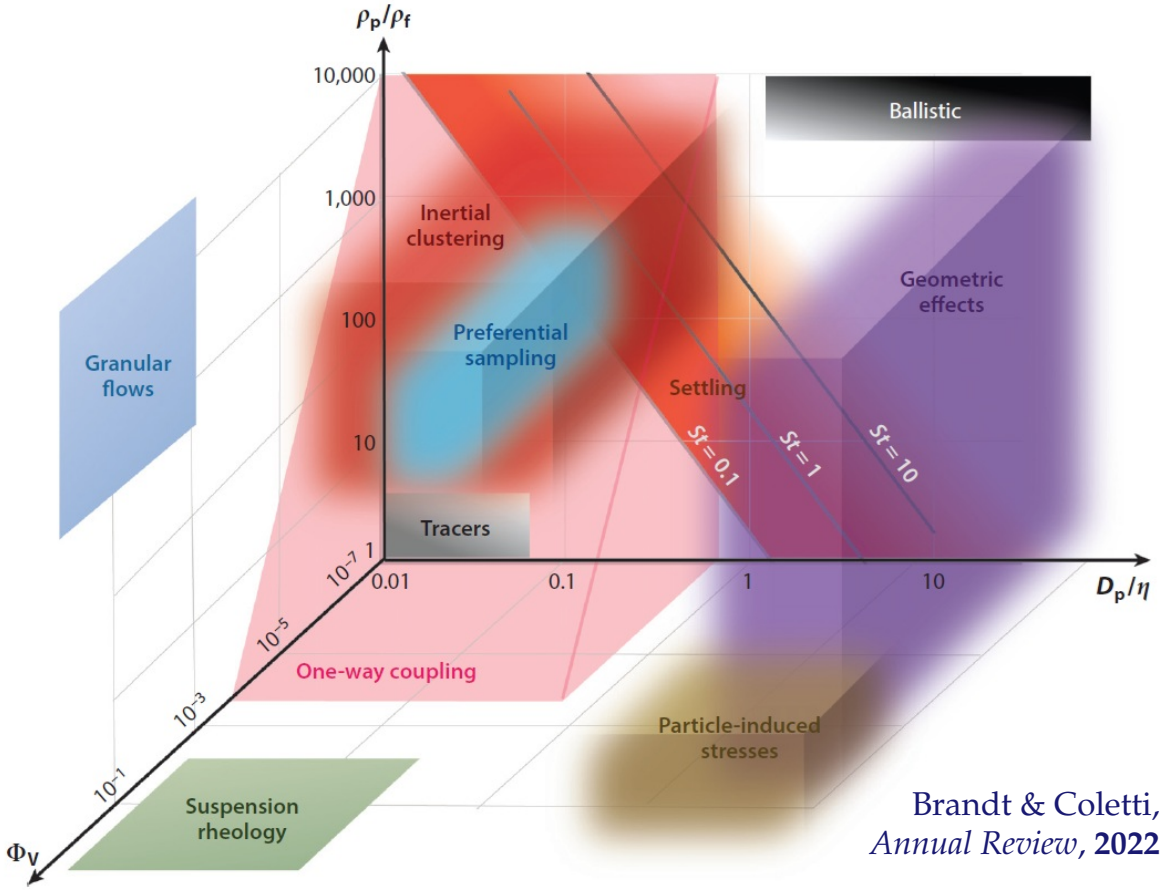
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Intertwined parameters

$\Longrightarrow$ complementary simulations and experiments



Brandt & Coletti, *Annual Review*, **2022**

Point particle approaches

- Central assumption : ***undisturbed flow velocity***
- Quasi-steady forces : drag (Stokes / nonlinear), lift
- Unsteady forces : added mass, pressure gradient, Basset
- Finite-size correction : Faxén : $\mathbf{u}_{eff} = \mathbf{u}(\mathbf{x}_p) + \frac{d_p^2}{24}\nabla^2\mathbf{u}(\mathbf{x}_p)$
- Validity : $d_p \ll \eta$, $Re_p \ll 1$

Particle equation (Maxey-Riley-Gattignol) :

$$\rho_p \frac{\partial v_i}{\partial t} = \overbrace{(\rho_p - \rho_f)g_i}^{\text{gravity}} + \overbrace{\frac{\partial \tau_{ij}}{\partial x_j} - \frac{\partial p}{\partial x_i}}^{\text{undisturbed flow}} + \overbrace{\frac{9}{2}\frac{\mu f_D}{r_p^2}\left((u_i - v_i) + \frac{r_p^2}{6}\nabla^2 u_i\right)}^{\text{steady state drag}}$$

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Point particle approaches

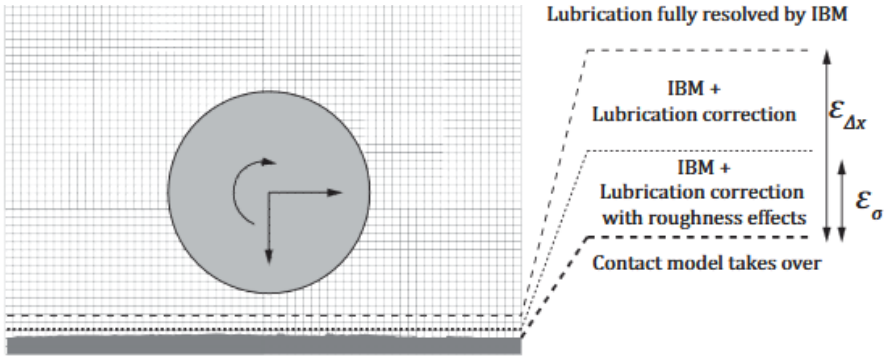
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Coupling :

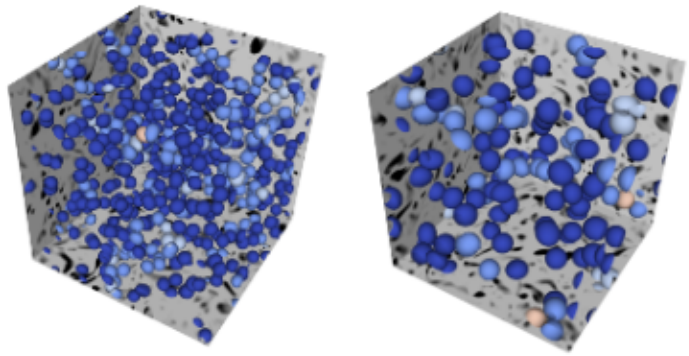
- 1-way coupling : fluid a priori + particle equation
- 2-way coupling : Navier-Stokes with particle back reaction
- ”4-way” coupling : inter particle momentum exchange
  - collisions
  - near field approaches
  - Lagrangian framework

Resolved particle approaches

- Finite size particles
- Key advantages :
  - exact surface stress
  - exact drag
  - no undisturbed velocity needed
- Key limitations :
  - computational cost
  - fewer particles
  - lower R_λ



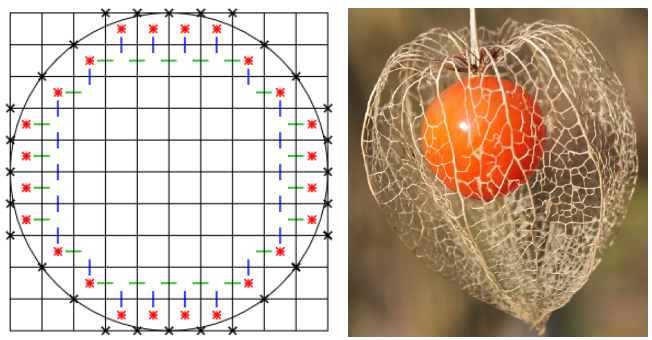
IBM : Costa et al, PRE, 2015



Basilisk : Boniou et al, JFM, 2025

A wide range of methods :

- Immersed Boundary Method
- Force Coupling Method
- Lattice Boltzmann
- Basilisk
- Physalis
- …

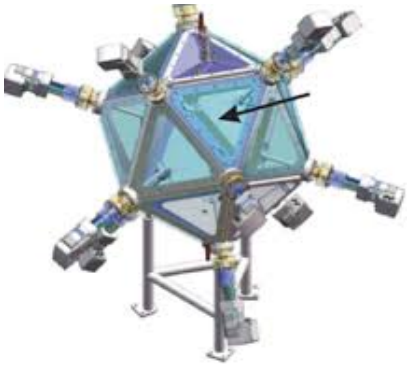


Physalis : Naso & Prosperetti, NJP, 2010

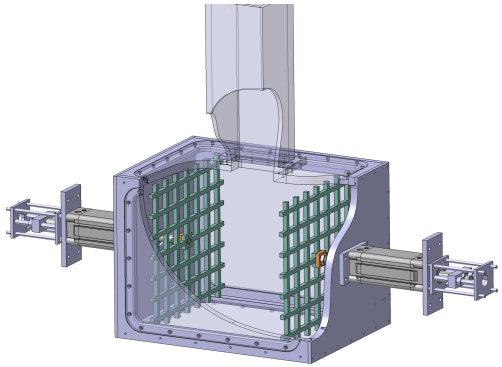
See review by Maxey, ARFM, 2017

Turbulent generators

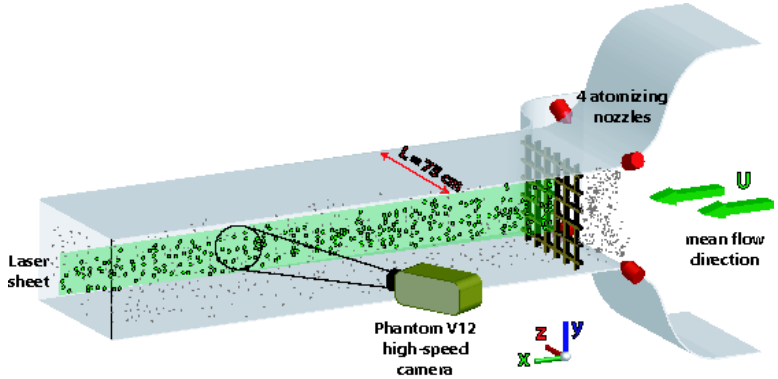
- Wind tunnels :
  - passive or active grids
  - *good homogeneity and isotropy*
  - *strong mean flow*
- Turbulent boxes :
  - von Kármán
  - polyhedra with fans / motors
  - oscillating grids
  - synthetic jets
  - *vanishing mean low*
  - *recirculation problem*
- In situ measurements



LEM : Zimmerman et al, RSI, 2010



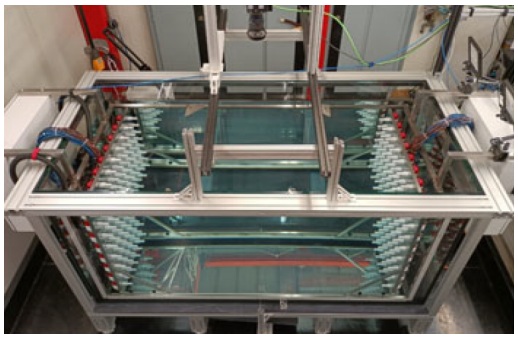
Oscillating grids : ENSTA



Wind tunnel : Monchaux et al, POF, 2009

Trade off :

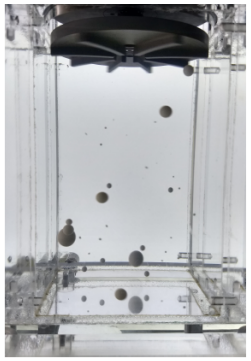
- Homogenity and sotropy
- Turbulence level R_λ

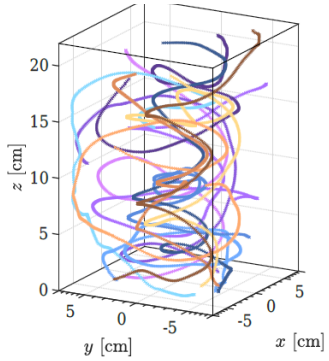


Jet arrays : Coletti group, Zurich

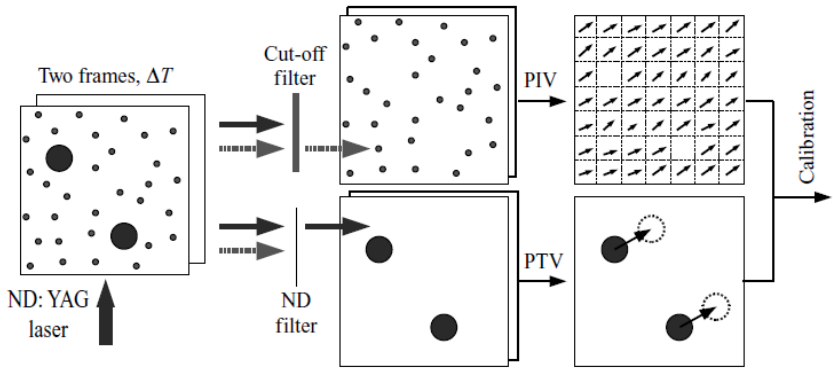
Turbulence measurements

- 2D : PIV, PTV
- 3D : Tomo-PIV, Holographic PIV, 3D-PTV
- Simultaneous fluid+particle :
  - fluorescence separation
  - refractive index matching
  - scale separation
- Opaque flows :
  - X-ray PIV
  - acoustic Doppler





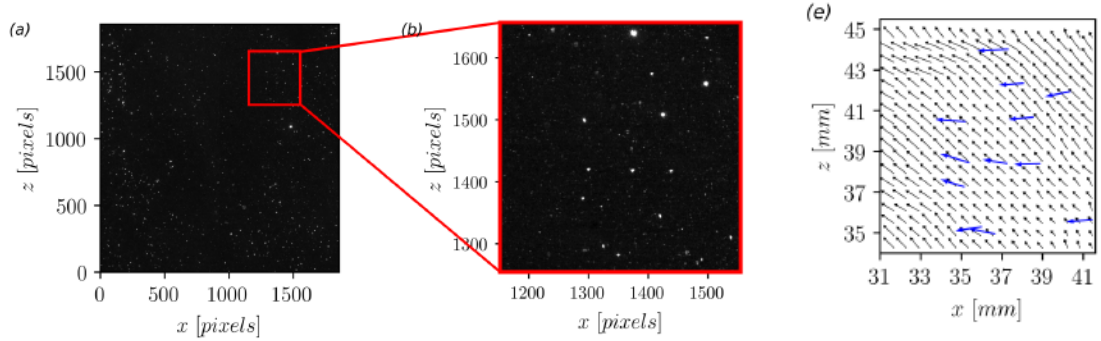
ENS Lyon group



Poelma et al, JFM, 2007

Seeding :

- Particle characterization
- Polydispersity
- Seeding homogeneity
- Disentangling parameters



Coletti group, Zurich